Digital Platforms as Data Vendors

Evidence from a Large E-commerce Platform

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Abstract

This paper studies a large e-commerce platform that has offered and commercialized seller-facing data analytics products. These products provide detailed information on customer sources and demographics as well as aggregate demand of a seller's product category. We find that sellers, especially small ones, are incredibly sensitive to the cost of data acquisition. This is despite its positive revenue impact and observed influence on positioning, marketing, and pricing decisions. Our analyses rely on high-frequency event studies as sellers adopted and used data analytics products, combined with quasi-experimental variation in the cost of data acquisition. A counterfactual simulation shows that a uniform reduction in the cost of data raises overall platform sales while reducing market concentration on the margin.

Introduction

Digital platforms such as Amazon, Google, and Alibaba have grown dramatically over the last two decades. They have also amassed an enormous amount of proprietary data, both a byproduct of their rapid expansion and a critical engine for their matchmaking capabilities that support further growth. This self-reinforcing dynamic has attracted regulatory concerns over its competitive implications among competing platforms as well as for the markets they intermediate (Kim 2019; Zhang et al. 2022; Bergemann and Bonatti 2023). However, recent studies have shown that some e-commerce platforms were able to boost seller revenue through data sharing (Berman and Israeli 2022; Bar-Gill et al. 2023). However, an open question remains on whether such data sharing can be commercialized. In reality, many platforms have become data vendors, hoping to capture economies of scope in their data and analytics capabilities while reinforcing preexisting network effects. Empirical evidence on such data transactions is sparse.

In this paper, we focus on a large e-commerce platform and study its experience offering and commercializing seller-facing data analytics products. Specifically, these products include historical and real-time information on a seller's customer sources and demographics as well as the aggregate demand in the seller's product categories. Most of the information provided is proprietary to the platform. Firms can gain access by purchasing annual subscriptions.

We first perform event studies to understand the return to data analytics for sellers. Here, we zoom in on sellers when they first adopted data analytics as well as when they acquired specific types of data. We can infer the latter using log data that tracked sellers' browsing patterns when accessing the data analytics product. We then use quasi-experimental variations in the pricing and bundling of data to gauge sellers' ex-ante willingness to pay for such data. Lastly, with the help of a demand model that incorporates

bundling, we conduct local counterfactuals to understand the optimal pricing and subsidies that the platform should charge for data.

The paper first introduces the e-commerce platform and the sellers operating on it. We then describe the data analytics product offered by the platform: the kinds of proprietary data that are offered, and the way they are presented and sold to sellers. Next, we narrow our focus down to a set of electronics and appliances product categories that have a high level of e-commerce penetration.[1]

We document three key reduced-form facts. (1) The adoption of both free and costly data analytics has increased steadily. (2) Sellers, especially small ones, were very sensitive to the cost of data analytics despite experiencing high returns. Costly data acquisition was thus concentrated among large incumbents. (3) Costly data acquisition led to significant changes in product positioning, marketing actions, and pricing.

In the third and fourth sections, we propose a demand model to synthesize the reduced-form facts and connect sellers' data acquisition decisions with their strategies and growth. We, again, find that small sellers were much more price sensitive, but conditional on size, young sellers are less price sensitive. Across various types of information, aggregate demand information primarily facilitated growth among large incumbents, while competitor information seemed particularly useful for young firms.

With the estimates from the structural model, we can conduct counterfactual simulations. Without taking a stand on the objective function of the platform, we only trace out the marginal impact of changing the cost of data on overall sales across all sellers on the platform and on concentration. We find that a uniform reduction in the cost of acquiring data increases overall sales on the platform while reducing industry concentration on the margin.

[1] 47% of all electronics and appliances are purchased online in China. See https://www.pwccn.com/en/retail-and-consumer/publications/global-consumer-insights-survey-2018-china-report.

Related Literature We are related to studies on the value of information and information technology, particularly among small businesses and in developing country marketplaces. Beyond those mentioned in the first paragraph, there is a large literature focusing on the impact of information and communications technology (ICT) infrastructure facilitated by the Internet (Cole and Fernando 2012; Hjort and Poulsen 2019). Studies on the role of information often focus on exogenous shocks to the availability of public information that is accessible to all businesses and consumers (Jensen 2007; Bai 2018). Our study contributes to the literature by modeling firms' data acquisition decisions.

We also contribute to an emerging literature on endogenous information acquisition and information design in industrial organization and quantitative marketing (Jin 2005; Dubé et al. 2017; Bergemann and Morris 2019; Jin and Vasserman 2019). The crucial difference is that we do not model demand or firms' competitive environment, nor do we make any structural assumptions about their priors along specific dimensions of knowledge that may be profit-relevant. Instead, we look at *marginal* changes to firm strategy, growth, and market concentration induced by endogenous information collection. Some of our findings, such as the effectiveness of product title adjustments as a marketing strategy, also echoes empirical evidence from in the digital marketing literature about the importance of search frictions, consumer salience, and obfuscation (Ellison and Fisher Ellison 2005; Ellison and Ellison 2009; Blake et al. 2018).

Third, we are related to an emerging literature on behavioral economics among firms, in which firms depart from the "profit-maximizing paradigm" (Armstrong and Huck 2010). Sellers are shown to exhibit large price sensitivity to the cost of information despite its large average revenue benefits. If we view data analytics as a new technology, our findings also expand prior studies on the resistance to beneficial new technology, particularly among small firms (Foster and Rosenzweig 1995; Conley and Udry 2010; Hanna et al. 2014; Atkin et al. 2017).

1 Background and Data

1.1 E-commerce Platform

Our investigation relies on data from a large e-commerce platform in China. Our study focuses on "Consumer-to-Consumer" sellers that are operated by individual entrepreneurs, which made up the vast majority of sellers online and more than 50% of total sales on the platform. Table 1 provides key summary statistics for these sellers. Specifically,

Table 1: Summary Statistics of sellers on the E-commerce Platform

C2C Seller Share	96%	C2C Sales Share	55%
Avg. Age (years)	4.1	Annual Sales (Median normalized to R)	
Female Owner	0.45	Mean	14.0R
Region: Coastal South	0.43	90th Pct.	24.3R
Region: West	0.12	Survival Dynamic*	
Median # Items Listed	15	After 3 Months	54%
Avg. Rating (out of 15)	6.3	After 12 Months	24%

Note: summary stats calculated based on a sample of active sellers of a randomly selected day in 2018. Survival probability was calculated with all newly registered sellers in 2016 and 2017 and conditioned on having stayed in the market for at least one month.

For every seller, we observe a rich set of observable characteristics. These include operating records such as seller registration date and type as well as lagged performances. We observe 138 primary product categories (or "industries"), with the largest firm in each getting an average of 9.63% market share and the top 20 firms getting an average of 31.6% market share. For our main analyses, we focus on several consumer electronics product categories for simplicity.[2]

[2]The product categories in our main dataset include "flash drive and storage", "DIY computer", "com-

1.2 Data Analytics Product

Sellers on the platform can acquire data from a set of proprietary data analytics products developed and offered by the platform. These products were first introduced in 2015 under a uniform and intuitive user interface accessible through the web. During our research window, it included information on the sources and characteristics of a seller's customers and visitors, historical and real-time aggregate demand for the entire product category, and competitor strategies. This information is mostly proprietary to the platform and divided into several modules as shown in Table 2.

Table 2: Modules Included in the Data Analytics Tool

Module	*Information Provided*	*Price/yr (RMB)*
Basic	Basic store-level information	Free
Traffic	Detailed traffic sources, conversion analysis, and customer demographics	888
Market Std.	Aggregate demand and trends top keywords for SEO	900 - 1188

Note: Price range indicate variation over time.

The platform offered a *basic* module for free. The platform used this basic version to communicate some crucial business statistics to sellers and to provide tutorials for data analytics. It contained basic statistics such as a seller's traffic, sales, and transaction statistics. If a seller opted in, it would gain access to the web and mobile interfaces that hosted all

puter hardware, peripherals, and monitors", "electronics components", "3C electronics accessories (Consumer electronics, Communications, and Computers)", "digital camera", "kitchen electronic appliances", "AI equipment", "MP3/MP4/iPod/recorder", "network and internet equipment", "home electronic devices", "video and audio electronics", "general electronics and electrical devices".

data analytics modules. However, below, we introduce several paid modules that would only appear after the seller has purchased them. Examples of these modules and the information they offer are shown in Figures A.1, A.2, and A.3.

The *traffic* module provided detailed analyses of the composition of visitors (traffic) and customers that a seller has attracted in the past. sellers could segment their customers by several observable characteristics to understand how—through what channel—they first discovered the seller and their browsing activities within the seller's website.

The *market* modules included information of the sellers' respective industries. Sellers' product category affiliation was defined by the platform based on minimum sales and product offering requirements in each product category. A seller hoping to purchase the market modules in a particular product category must first have offered a product with non-trivial sales in that category. The *standard market* module showed trends in aggregate demand both in terms of realized transactions and in terms of top keywords searched. The trends were shown in real time and for historical periods up to one year. The *pro market* module extends the historical records to three years and includes additional information on the top-selling and top-searched sellers, brands, and products. The pro module costed about 10 times as much as the standard version and thus had trivial adoption rate, we thus focus on the standard module.

2 Reduced-Form Evidence

In this section, we document three sets of descriptive and reduced-form facts. Our goal is to provide direct evidence on sellers' propensity to acquire data as well as their impact on seller strategies and outcomes.

2.1 Data Analytics Adoption

Fact 1: the prevalence of data acquisition has grown steadily but has remained concentrated among large incumbent sellers We first describe the acquisition pattern of different data modules across seller size groups. In Figure 1, we use an area plot to demonstrate the percentage of sellers who used and who paid for data analytics, which corresponds to the axis on the left. The dark blue areas indicate the portion of sellers, at a given time, that had never activated the basic (free) version of the data analytics product.

Based on the platform's own definition as well as our categorization, we omit sellers below the 70th percentile, which had negligible revenues or costly data acquisition. The remaining sellers are defined as "active" sellers. A majority of them had activated the basic

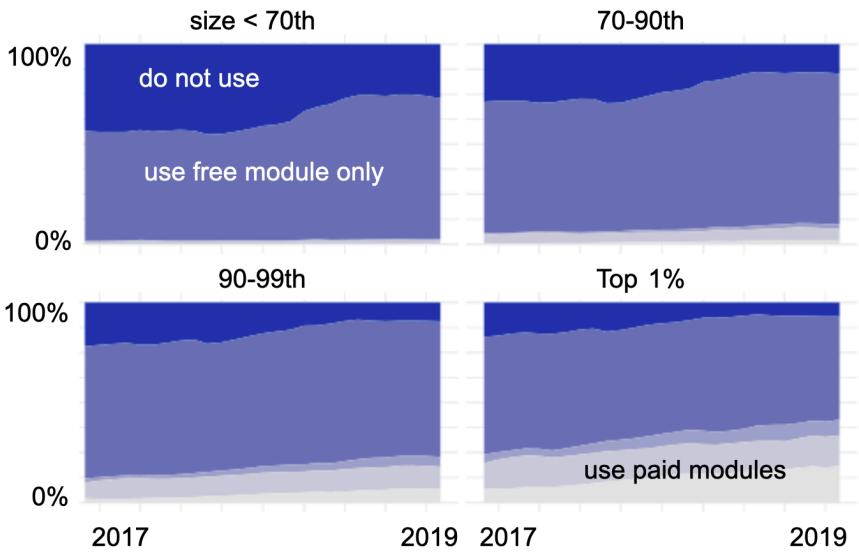

Figure 1: Adoption Patterns across Modules, Size Groups, and Time

Notes: Adoption patterns of B2C sellers by seller size over time. Adoption is measured by activation of the corresponding data module. The four quadrants represent seller size groups measured as sales in the previous month. Based on the platform's own definition as well as our categorization, we omit sellers below the 70th percentile, which had negligible revenues or costly data acquisition. We categorize sellers in the 70th to 90th percentile as small-sized, 90th to 99th percentile as medium sellers, and the top 1 percent as large sellers. The solidly dark areas represent sellers who had never activated the basic (free) version of the data analytics product. The shaded area measures those that had but never purchased any other modules. The lightly shaded are represents sellers that purchased at least one data module.

version during our research window, indicating a growing awareness of data acquisition. However, costly acquisitions remained rare in comparison. Nonetheless, based on changes in the popularity of the paid modules (information on detailed traffic breakdown for own seller and that on the sellers' respective market), sellers of all sizes seem to have become increasingly likely to pay for data over time.

Looking at each of the two paid data modules, as shown in Figure 2, the adoption patterns seem to be consistent: linear growth in adoption rates (blue line, right axis) with concentration among large incumbents (red line, left axis). Sellers were particularly interested in obtaining market information, although a proper evaluation of sellers' valuation of such information must account for price differentials and seller heterogeneity, which we turn to next.

Figure 2: Adoption Patterns across Modules Over Time

Notes: The graph shows the penetration rate of each of the two paid data modules in terms of number of sellers and market share of those sellers among all active sellers over time.

2.2 Seller Sensitivity to Cost of Information

Fact 2: sellers, especially small ones, were very sensitive to the cost of acquiring information We now take advantage of a sharp back-end price change in the market module to understand sellers' sensitivity to the cost of the information modules.

The black line in Figure 3 plots the pricing of the standard market module offered in the data analytics tool, corresponding to the right axis. The red line plots the amount of first-time purchases of the corresponding module, corresponding to the left axis. There was obvious seasonality in consumers' purchase decisions.[3] Nonetheless, quantity fell in response to each price hike. The same pattern is true for sellers' renewal decisions, which

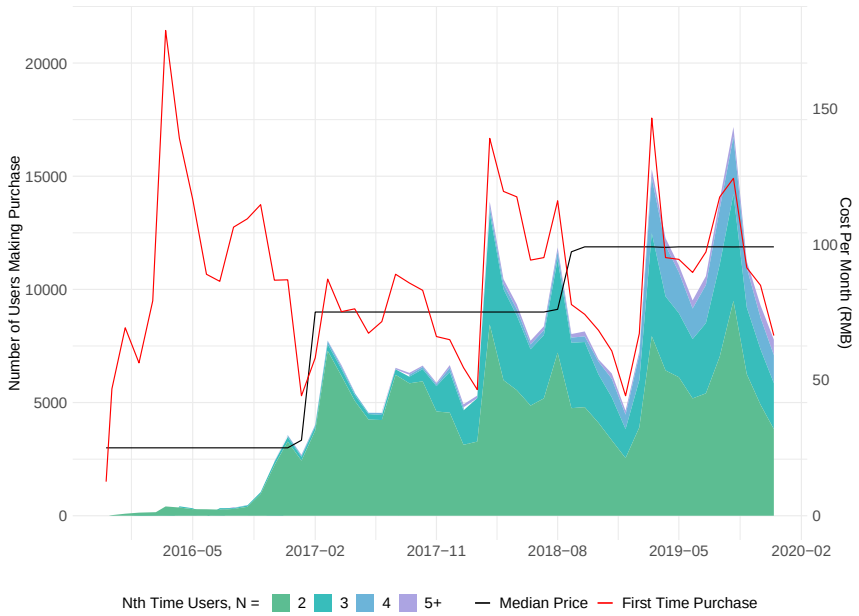

Figure 3: Adoption and Price Change of Market (Std) Module

Notes: The number of orders placed for the *market* (standard) module. The Red line depicts the number of orders placed by first-time users, while the colored area represents the renewal purchases, which are broken down by the "Nth Time Users"—the number of times a customer has purchased the module in prior years. The number of users is re-scaled.

[3]Seasonality is driven partially by major promotion events and shopping festivals on the platform. Sellers are most likely to adopt after the Spring Festival and during the months when they prepare for major shopping events.

are represented by the colored area graphs based on the number of renewals.

We now code the price hikes as an instrumental variable in order to quantify sellers' price sensitivity. Define instrument z for price p as follows, where i indexes sellers and t calendar weeks:

$$z_{it} = \begin{cases} 1 & \text{if t >= price change date} \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

We consider the purchases made around the price change that occurred at the end of August 2018. During this episode, the platform increased the prices for both the market and traffic modules, citing reliability upgrades as the main reason.

The main specification is simply

$$\ln Price_{it} = \delta z_{it} + x'_{it}\beta + \varepsilon_{it} \tag{2}$$

$$Purchase_{\text{mkt},it} = \theta \ln \hat{Price}_{it} + x'_{it}\alpha + \varepsilon_{it} \tag{3}$$

Results are presented in Table 3 below. For the market module (columns 1-3) that contained aggregate demand data for transactions and search keywords, a 100 RMB increase in price would see adoption drop by 35.1%, implying a demand elasticity of -3.16. Notice that the dependent variable is the pre-period average adoption rate measured in the *per-week* purchase rate of the market module, at 0.05%. Moreover, the price sensitivity is substantially higher among small or new sellers. The results for the traffic module are similar to those for the market module.

Table 3: Demand Elasticity of Market Modules

	Dependent variable:					
	$Purchase_{market,it}$			$Purchase_{traffic,it}$		
	(1)	(2)	(3)	(4)	(5)	(6)
Price	-0.002***	-0.002***	-0.002***	-0.001***	-0.001***	-0.001***
	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Price × Small Seller		-0.001***			-0.0003***	
		(0.00003)			(0.00001)	
Price × New Sellers			-0.0004**			-0.0001**
			(0.0002)			(0.00004)
Dep. Var. Mean	0.005	0.005	0.005	0.003	0.003	0.003
R^2	0.086	0.086	0.086	0.047	0.047	0.047

Notes: These regressions are based on a random sample of about 52,000 sellers in consumer electronics product categories from 2017 to 2018. A seller is in the sample if she is *not* a current user of the corresponding modules. Prices for a 12-month subscription are log-transformed. The instrument used is an indicator for whether t is after the price change date and 0 otherwise. All regressions include sellers, year and month of year fixed effects as well as variables indicating current renewal cycle by module. Regressions with interactions use z_{it} and lagged indicator for being small/young as the first stage instrument for p_{it} and p_{it} interacting with the current period indicator. A seller is small if the current period sales are between the 70th and 90th percentile. A seller is young if its registration date is within a year of t. ***p<0.01.

2.3 Data Acquisition and Strategy

Fact 3: Costly information acquisition observably influenced positioning, marketing, and pricing decisions What do sellers do after acquiring costly data? To answer this question, we first focus on sellers who have purchased the data. Log data on the type of data they have visited at any given time allowed us to link it directly to subsequent strategy implementation. We conduct a simple regression where the dependent variables are strategies implemented, while the independent variables are visit histories of the different data modules during the same week. Specifically, let i index sellers and j the data modules, we

can conduct a horse-race regression as follows.

$$Strategy_{it} = \alpha_i + \alpha_t + \sum_{j \in J} \theta_j Visit_{ijt} + x'_{it}\beta + \varepsilon_{ij}$$

The estimated θ_j's are presented in Figure 4, in which different colors represent different dependent variables and product strategies. Specifically, this figure includes pages reserved for the Pro version of the data analytics modules which we omit in prior analyses due to low adoption rate. This is because visit log data is much higher frequency, allowing us to capture its relationship with subsequent seller strategies precisely.

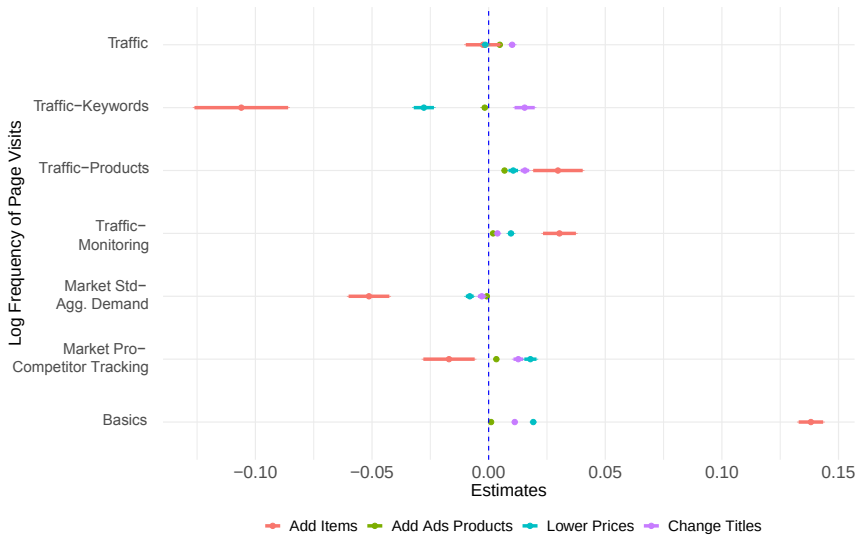

Figure 4: Strategy Implementation and Information Acquired

Notes: Sellers' interaction with the data analytics products is tracked by the platform. We calculate their site visit records over a sample of sellers in consumer electronics-related industries from March to August 2018 (prior to price changes and bundling of market modules). All strategies are measured in frequencies and are log-transformed. Visit patterns are captured by dummies indicating having visited any pages in the corresponding module. Additional control variables include the number of products listed, the current week's total sales and traffic. All regressions also include seller- and week-of-year-fixed effects.

Lastly, we conduct event studies around the purchase of market and traffic data modules to directly visualize how seller strategy changed dynamically immediately after ac-

quiring information.

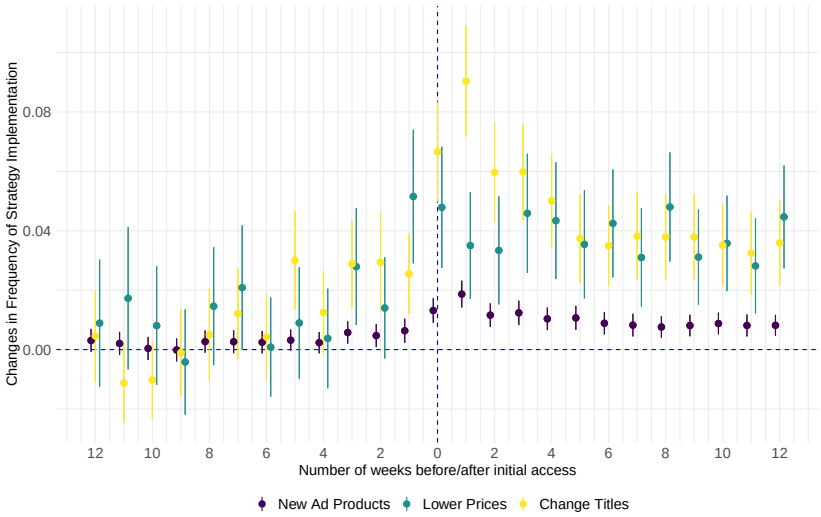

Figure 5: Strategy and Timing of Adoption: Market Info

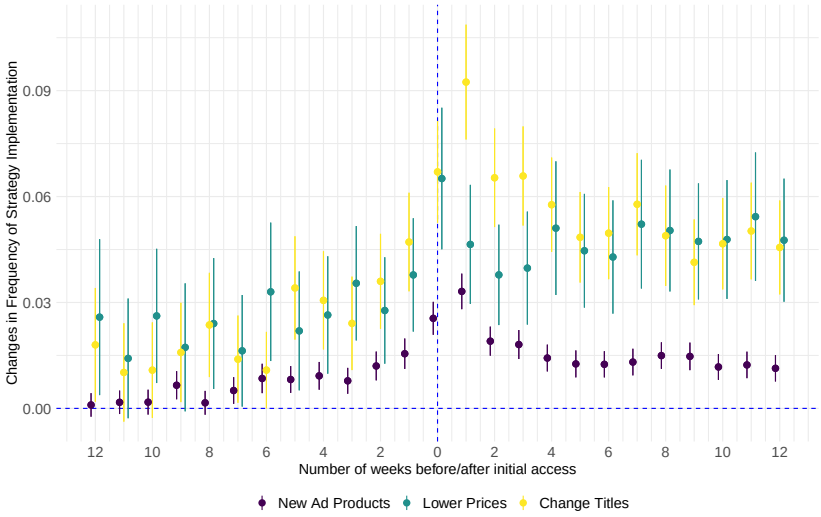

Figure 6: Strategy and Timing of Adoption: Own Traffic Info

*Notes:*sample consists of sellers in consumer electronics-related industries that have ever purchased a market module. Regressions include seller- and time-fixed effects. Each dot-bar represents coefficients on a dummy equal 1 if for the seller i current week t is j weeks before/after subscription to market module start. The omitted category is 13 or more weeks before the start. Dependent variables of frequency of employing certain product strategies measured in log.

Figures 5 and 6 demonstrate the frequency of product strategies implemented with a fixed effect regression, where we take all sellers who have purchased the market module,

and regress strategies implemented on time relative to adoption (i.e. a dummy equal 1 if a particular week t is k week before/after adoption for seller i) and seller as well as calendar week fixed effects.

$$Strategy_{it} = \alpha_i + \alpha_t + \sum_{j \in \{-12,\ldots 12\}} \theta_j RelativeWeek_{ijt} + RenewalCycle_{it}\beta + \varepsilon_{ij}$$

The results presented in Figures 5 and 6 confirm a positive correlation between the implementation of product strategies and the acquisition of costly information on market and own traffic. Sellers have higher frequencies of changing product titles, lowering prices, and adding new products after acquiring costly data. The demand side information that sellers purchase could inform these expansion-oriented strategies. The increases peak at the time of purchase and gradually level off over time.

2.4 Impact of Data Acquisition

Fact 4: Information acquisition facilitated revenue growth Does acquiring information actually make implementing strategies more effective, facilitating revenue growth?

An easy way to get at this effect is to use a matching estimator and look at the effect of information acquisition on revenue growth, which is a particularly important metric for small- and medium-sized businesses. To do so, we first construct a matching sample for the group of sellers that have purchased each data module. Specifically, we match on all observable characteristics of the seller, including size, age, and industry dummies, etc. We also match sellers based on their past sales and strategy performances. Please find details of the matching estimator in Appendix C.

We can then conduct the following regression on the matched sellers. Here, X includes variables used in the matching function for all eleven weeks prior to t_0: observable characteristics of the seller, actions, and performance. This means that the outcome variable at

time t_0 is included in X.

$$\ln Sales_{i,t_0+k} = \gamma_k Adopt_{i,t_0} + X'_{i,t_0}\beta + \alpha_k + \varepsilon_{k,i} \; \forall k = 1, ..., 11$$

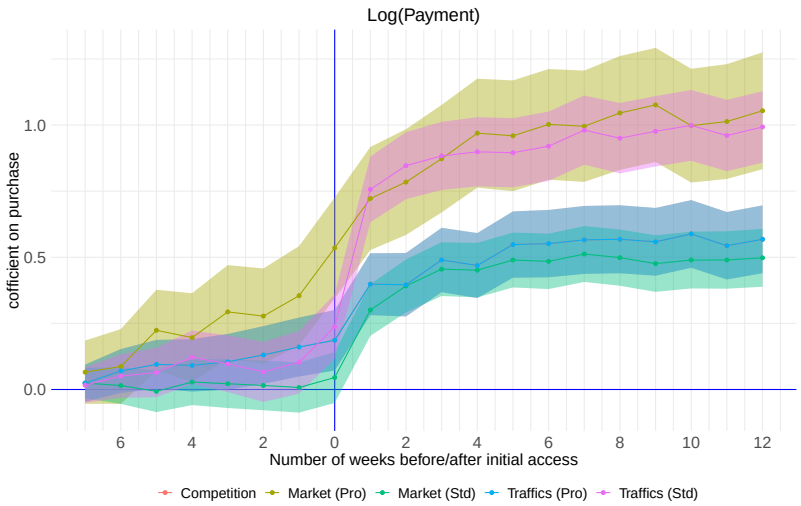

Figure 7: Effects of Data Acquisition on Log Sales

Notes: this graph shows coefficient γ_k for $k = -11, ..., 11$ where $k = 0$ is the week of adoption, separately for each information module. The sample size for each type of information varies and is determined by the number of users in the consumer electronics industry sample that have adopted the corresponding module for the first time between 2017 and 2018. Each matched sample is constructed independently following the procedure described in the appendix. The dependent variable is the sales in the previous week in log scale. All regressions include controls for seller characteristics as well as seller performance and action profiles in the pre-adoption period.

Figure 7 plots the implied treatment effect γ_k across time k on the horizontal axis, with different colors representing different types of information acquired. We can detect sizable treatment effects based on comparing sellers that have identical observable characteristics as well as performance and strategy profiles. Results from the matching estimator suggest that sellers who paid for the information indeed outperformed sellers that do not, especially since the modules being purchased are about markets and traffic. Moreover, as shown in Figures 8 A.5 and A.6, compared to large sellers, small sellers can achieve higher growth

(in log terms) relative to their matched counterparts.

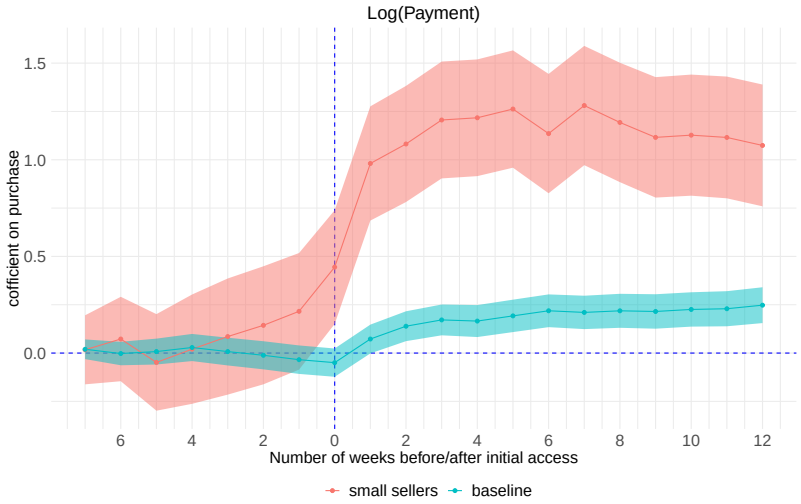

Figure 8: Effects of Acquiring Aggregate Demand Information on Log Sales, by seller Size

Notes: this graph repeats Figure 7 for the standard market only, but by seller size groups.

An Experiment on the Traffic Module The key weakness of our approach is that there may exist unobserved motivation/skill/cost differences across the matched and matching samples that endogenously determine information acquisition but are not captured by pre-period actions and outcomes.

To address the selection issue, we conducted an experiment during a promotional campaign for the traffic module implemented in June 2020. There, 50% of sellers who had never purchased any modules got a seven-day free trial for the traffic module.

The campaign launched on June 22, 2020. We track sellers for the two weeks before and four weeks after the campaign. 1.64% of treatment-group sellers took up the free trial, which was objectively low but 32 times higher than the control group. In addition, only 3% of sellers purchased the traffic module after trying it out. This reflects our focus on sellers who have never purchased any data and may thus have low purchase intent for such data. Nonetheless, we can use this experiment to examine the impacts of acquiring traffic data.

To do so, we run the following 2SLS specifications, in which data adoption is instrumented by treatment assignment. Let i index seller and t calendar day.

$$\text{1st Stage: } Trial_{it} = \gamma Post_t Treat_i + \alpha_i + \alpha_t + \varepsilon_{it} \tag{4}$$

$$\text{2nd Stage: } Y_{it} = \beta \hat{Trial}_{it} + \alpha_i + \alpha_t + \varepsilon_{it} \tag{5}$$

Table 4: Sellers' Performance Using Free Trial

	Dependent variable:				
	# Visitors	Conversion	Revenue	# Buyers	# Orders
	(1)	(2)	(3)	(4)	(5)
Free Trial	-0.317	0.086***	1.033	0.827**	0.837**
	(0.449)	(0.020)	(0.792)	(0.322)	(0.326)
Dep Var Mean	5.25	0.05	6.21	2.17	2.2
Observations	8,809,122	8,699,674	8,809,122	8,809,122	8,809,122
R^2	0.899	0.481	0.671	0.867	0.864
Adjusted R^2	0.897	0.468	0.663	0.863	0.861

Notes: Regressions based on a sample of 220k sellers who are sampled from all the targeted sellers of the promotion campaign. In this sample, 50% of sellers are randomly selected to receive the promotional message for the free trial. Observations are at seller-day level spanning from June 8 to July 20, 2020. The table presents the estimated effect of the free trial from the second stage specification of 4. All outcome variables are in log scale, except for the conversion rate, which is measured as the share of visitors who placed orders on that day. **p<0.05, ***p<0.01.

Table 4 presents the estimated seller performance results using the IV specification. Taking up the free trial has no significant impact on sellers' daily revenues (measured in log scale) for the subsequent month[4]. Upon decomposing the revenues, we found that while free trial sellers did not attract more site visitors, they had higher conversion rates and served more consumers. One possible explanation for such a pattern is that with

[4]However, sellers taking up the free trials earn higher revenues when measured in levels. The results are mostly robust to different winsorization thresholds, presented in table A.1.

better information about traffic sources, free trial users improve the efficiency of their marketing activities. Even though they did not attract more visitors, those visitors these sellers managed to attract are more likely to become actual buyers.

3 Model: Data Acquisition and Strategy

Following the evidence presented in the section above, we propose a model of information acquisition and product strategy. In order to understand firms' decision to acquire information together with their desire to implement product strategies, we must estimate two types of key parameters. First, stores' valuation of different types of information in relations to each product strategy (γ). Second, stores' costs of implementing product strategies (c).

Let i index stores, j DAP (data analytics product) module (bundles), t time period (week), and k information that can be bundled into DAP modules. Further denote the characteristics of a store as x and its renewal count as rc. On the other hand, denote stores' product strategy as s. Our focus here are the number of times new products are introduced or existing product re-positioned through title changes.

The most important primitives of the model is stores' valuation of how different pieces of information can lead to increasing effectiveness in executing product strategies, as reflected by bringing in additional sales growth. We model these valuations and correlations directly. For store i at time t, conditional on the stream of product strategies that it would

implement, $\mathbf{s}_{it}$, its valuation of each DAP module j would be as follows.

$$v_{ijt}(\mathbf{s}_{it}) = \mathbb{E}\left[\sum_{s}\sum_{k\in K_j, t'\in T_j} \gamma_{s,ikt'} \cdot s_{it'}\right] \tag{6}$$

$$\text{where } \gamma_{s,ikt} = \gamma_s(x^{\gamma}_{it}, \mathbf{1}_{yr}, \mathbf{1}_{week}, \mathbf{1}_k; \theta_\gamma) + \varepsilon_{s,ikt} \tag{7}$$

$$\varepsilon_{s,ikt} \sim \mathbf{N}(\mathbf{0}, \Sigma_\gamma(\sigma_\gamma, \rho_{\gamma,t}, \Sigma_k))] \tag{8}$$

Here, x^{γ} represents observable characteristics of the store, including firm age, size, business type, and industry dummies. 1_{week}, 1_{yr}, and 1_k are fixed effect for week of year, year, and content. Lastly, as in Equation 8, γ is allowed to vary based on an unobservable component ε_s. It is assumed to be i.i.d. across stores i and follows a joint normal distribution Σ_γ. Σ_γ is parameterized so that ε_s are correlated across content types k based on Σ_k, and across time based with serial correlation term $\rho_{\gamma,t}$. Notice that, in this model, we assume that γ_s's are independent across different sets of strategies.

Based on the effectiveness of acquiring information, we can specify a simple logit models to understand stores' choices to acquire information and to implement product strategies.

$$u_{ijt}(\mathbf{s}_{it}) = v_{ijt}(\mathbf{s}_{it}|\gamma) + \alpha_{it} p_{ijt} + \xi_j + \varepsilon_{ijt} \tag{9}$$

$$\text{where } \alpha_{it} = \alpha_0 + \alpha_1 x^{\alpha}_{it} \tag{10}$$

$$\text{and } \varepsilon_{ijt} \sim EV1(0, \frac{\pi^2}{6}) \tag{11}$$

$$\text{so that } \Pr(\mathbf{1}_{ijt}|\mathbf{s}_{it}) = \Pr\left(u_{ijt}(\mathbf{s}_{it}) = \max_{j'}\left[u_{ij't}(\mathbf{s}_{it})\right]\right) \tag{12}$$

Here, p_{ijt} denotes the price for the information module j. Stores' information purchase decisions are intertwined with their desire to implement product strategies s, taking into account their valuation of the complementarity between information acquisition and strategy

implementation, γ. Let $\mathbf{s}_{it} = (s_{it}, ..., s_{i,t+12})$ and each s_{it} be a binary indicator for strategy implementation.

$$\Pr(s_{it} = 1 | \{\mathbf{1}_{ijt}\}_J) = \Lambda \left[\sum_j \mathbf{1}_{ijt} \cdot v_{ijt}(s_{it}|\gamma) + c_{s,it} \geq 0 \right] \tag{13}$$

$$\text{where } c_{s,it} = x_{it}^{c\prime} \theta_c \tag{14}$$

Similar to Equations 7 and 8, we specify c as a random effect that denotes the unobserved baseline net gain of implementing strategies s. We assume that c_s's are independent across different sets of strategies and that they are only correlated with γ_s's based on observables. Notice that we do not consider the cost of information acquisition p here. This is because that cost would be sunk once the store has made the acquisition decision, therefore, any subsequent strategy decision should not depend on it.

Lastly, assume that store revenue follows a log-normal process in which the mean depends on observable characteristics and lagged sales, as well as the expected gain from implementing strategies. For our baseline version, we assume that stores are, on average, rational in their expectation of the impact of information acquisition.

$$r_{it} \sim N(\mu_{r,it;\theta_r}, \sigma_r) \tag{15}$$

$$\text{where } \mu_{r,it} = x_{it}^{r\prime} \theta_r + \sum_j \mathbf{1}_{ijt} \cdot v_{ijt}(\mathbf{s}_{it}|\gamma) \tag{16}$$

For simplicity, we start with one $\mathbf{s}_{it}'$, product title change, and three type of information, aggregate demand, top stores/product/brands, and competition. This reduces the computational burden.

4 Model Estimation, Results, and Counterfactuals

4.1 Model Estimation

We estimate the model with a simulated maximum likelihood approach focusing on the likelihood of streams of (1) product strategy implementation: adding products and changing product titles; (2) acquisition of market and competition information; and (3) sales outcome.

For each set of parameter proposals, we generate 50 pseudo-random draws of $\varepsilon_{s,ijt}$ and compute the likelihoods. Averaging over $\varepsilon_{s,ijt}$ for the corresponding module j gives us a set of v_{ijt}'s, which we can directly use to calculate likelihoods in equations 12, 13, and 15.

4.2 Identification

We estimate the model on a subsample of 500 stores and 18 periods are reported in the following section while bootstrapping for standard error. The time subset is chosen to include both pre- and post-price change in our instrumental variable regression in Equation 1. During the price change, competition information is merged into the market pro module, while the competition module becomes unavailable.

First, we can compare sales outcome and information acquisition decisions across and within stores, conditional on observables and lagged performance. Our setting also allows us to do so under different pricing and contract space environments. This helps us pin down the scale of γ's and α's, conditional on the observed strategy streams $\mathbf{s}_{it}$. How γ's change over time and correlates across different content types k allows us to fit the linear model in equation 7 and obtain θ_γ, σ_γ, $\rho_{\gamma,t}$, and Σ_k.

Conditional on the effect of acquiring data and the realized information acquisition decisions ($\mathbf{1}_{ijt}$), we can then maximize the likelihood of observing the strategy stream for

each store. Ideally, we would want to observe situations where information acquisition would not be possible, therefore reducing the likelihood that acquiring information would make a difference. But given that the v's are already identified above, the observed frequency of strategy alone is enough to pin down the levels of c and therefore the θ_c's

4.3 Estimation Results

Estimation results are presented in Tables B.1.

Our price sensitivity estimates capture the same pattern across store size groups as in the reduced-form section above. Log store size, as measured by lag revenue, strongly and negatively contributes to the magnitude of overall price sensitivity. Conditional on store size, however, older stores are actually more price-sensitive than newer ones. On the other hand, γ, the benefit of acquiring market information is higher for older stores, while newer stores benefit more from acquiring competitor information. ξ terms capture the premium on each module or information piece unexplained by revenue impact (γ), relative to the market pro module and the top rankings information. The expensive market pro module seems to command a premium over its cheaper counterparts, which can explain why its purchase share is not small compared to the other two modules, despite significantly higher prices. However, the content contained in that module, the top rankings (of stores, products, and brands), actually have an inferior baseline revenue impact.

Using these hyper-parameters, we can calculate latent parameter distributions, reported in Table 5. There is large variations in price sensitivity, as identified by differential adoption responsiveness to cost changes in data acquisition over time across store size and age groups.

In light of the magnitude of α's, the estimates of γ parameters indicate that the effect of information acquisition, which is estimated based on actual sales performance, can be very large. However, there is significant heterogeneity across stores.

Table 5: Latent Parameter Distributions

Statistic	Mean	St. Dev.	Min	Pctl(25)	Median	Pctl(75)	Max
α(size,time)(e-4)	-9.09	5.6	-21.97	-14.86	-7.84	-4.78	1.18
$\gamma_{\text{top rankings}}$	0.42	2.36	-5.07	-1.27	0.40	1.96	20.41
$\gamma_{\text{agg demand}}$	1.66	1.62	-13.72	0.59	1.60	2.68	6.33
$\gamma_{\text{competitors}}$	2.51	2.14	-5.32	1.01	2.69	4.17	7.57

We can further dissect the sample of stores by size and age to examine the heterogeneity in the impact of data on sales. Figure 9 shows our results with dissection across firm size. It demonstrates that although the median firm in both size groups has a positive valuation for all types of information, there exists significant heterogeneity within each size group. Meanwhile, comparing median differences across size groups, supply (competitor) information and traffic details (own customer sources and demographics) seem to be particularly useful among small stores. Only aggregate demand information is more highly valued among larger stores.

Figure 9: Effects of Information Acquisition on Log Sales across Data Type and Size Groups

Notes: this is a box plot of the value of information, measured in $\hat{\gamma}$ estimate from our model across information type and across store size group (50-50 dissection).

Figure 10 repeats the above analysis but dissects stores based on age. It confirms and

amplifies three patterns that we have seen above in the comparison across firm size (firm size and age are highly correlated). Aggregate demand information is very valuable for incumbents while being virtually useless for the average young firm.

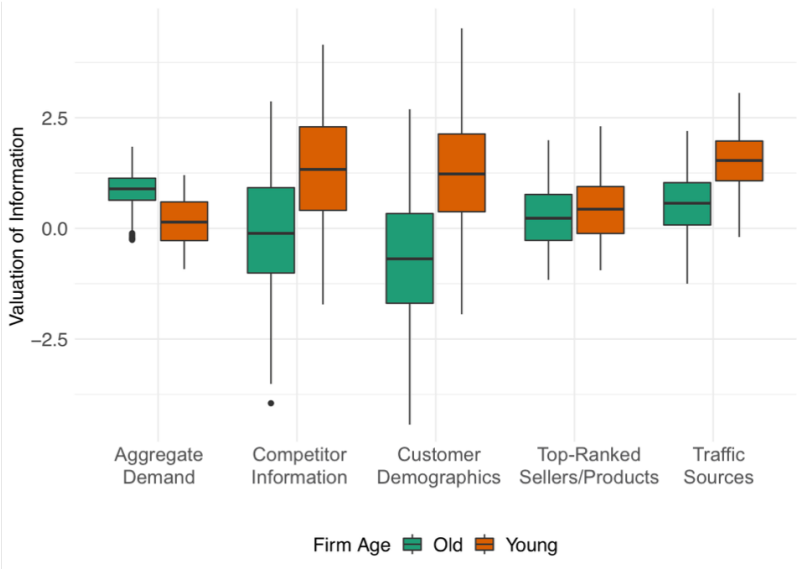

Figure 10: Effects of Information Acquisition on Log Sales across Data Type and Age Groups

Notes: this is a box plot of the value of information, measured in $\hat{\gamma}$ estimate from our model across information type and across store size group (50-50 dissection).

4.4 Counterfactual Data Pricing: Marginal Changes in Overall Growth and Concentration

With the demand estimates, we can conduct simple counterfactual simulations that alter the pricing of the data product. The outcome that we focus on is the marginal impact on overall growth in sales across all stores and that on market concentration, as measured by the Herfindahl-Hirschman Index (HHI) among the stores in our sample.

Figure 11 plots the simulated outcomes, where the color gradient of each dot represents the intensity of counterfactual discounts applied uniformly across all data modules and for

all stores. Darker means deeper discount. On the vertical axis, we show marginal changes in overall sales growth across all stores, while the horizontal axis shows the marginal change in market concentration.

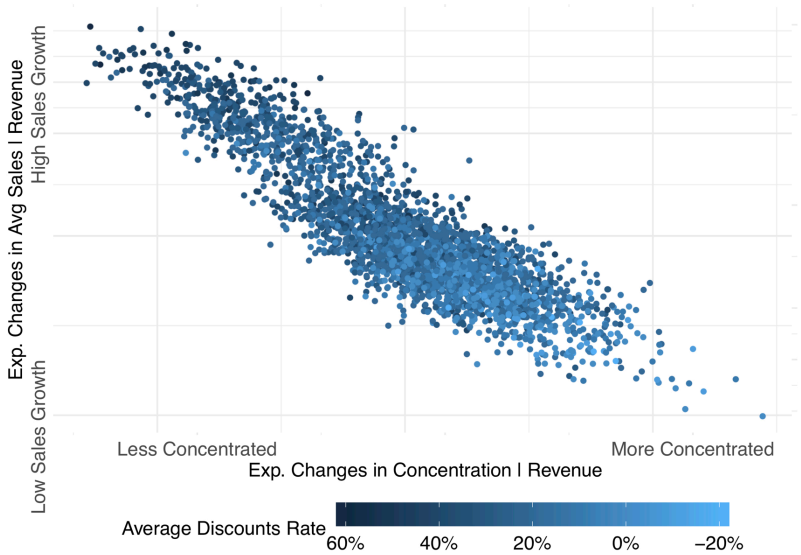

Figure 11: Effects of Data Acquisition on Log Sales

Notes: this graph shows the marginal impact on overall sales across all stores (vertical axis), and on industry concentration (horizontal axis) as measured by the HHI index when the platform applies uniform discounts on all data analytics products.

Raising the cost of data depresses overall sales growth, and it does so by disproportionally suppressing data acquisition among small stores, therefore increasing concentration. Sales growth among large firms brings in immediate revenue for the platform, while sales growth among small stores may beget larger gains in future growth of the platform. The platform can therefore balance these forces to determine the profit-maximizing prices on different types of data.

5 Conclusion

In this paper, we acquire detailed proprietary data about online stores on a large e-commerce platform. They allow us to uncover several key facts about how different firms value different types of data, the strategies that are influenced by those data, as well as the ultimate growth impact.

The implication of our work goes beyond the online retail setting. When information is costly as opposed to freely available in the market (credit history data of borrowers can be easily accessed by almost all accredited lenders and insurers in the US, for example), our results imply that large firms are much more likely to acquire information than their smaller counterparts. We also demonstrate that information acquisition *can* lead to significant sales growth, often through non-price channels, which may exacerbate market concentration. To mitigate this endogenous force towards concentration, a social planner can reduce the cost of information, particularly among small and young firms and for information on their own customers and operations as well as on their competitors.

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Appendix

A Additional Tables and Graphs

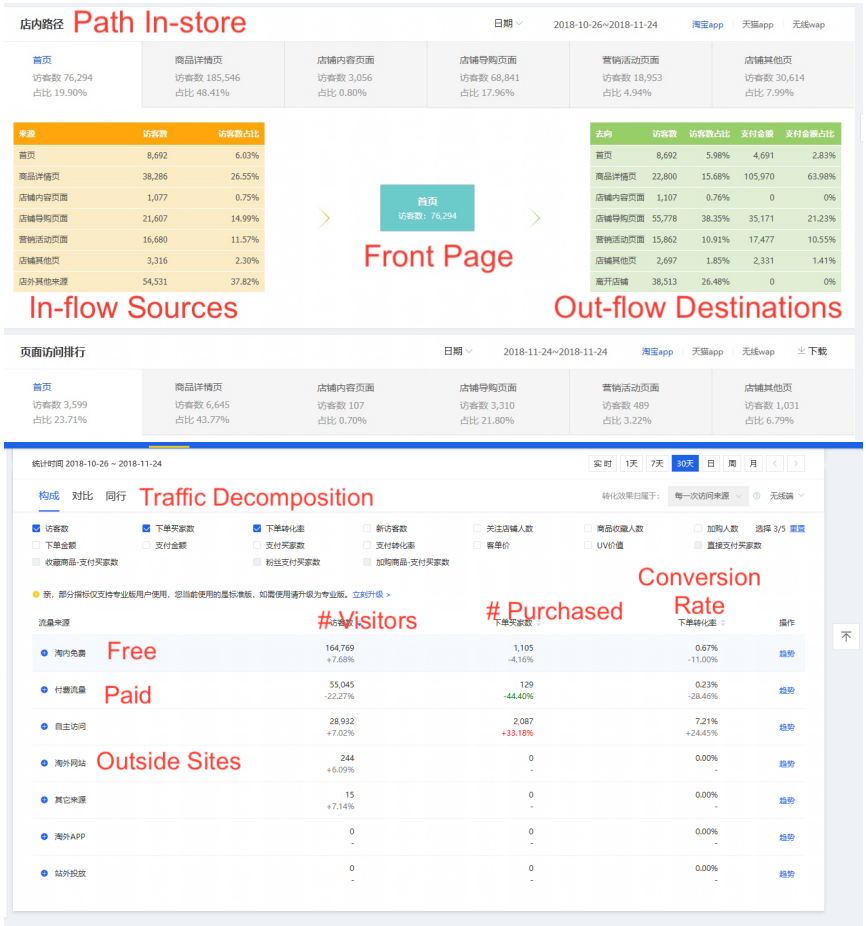

Figure A.1: Traffic Module Interface

Notes: This is a screenshot that illustrates the web interfaces of the paid traffic module in the data analytics tool.

Figure A.2: Traffic Module Interface

Notes: This is a screenshot that illustrates the web interfaces of the paid market modules (standard and pro) in the data analytics tool.

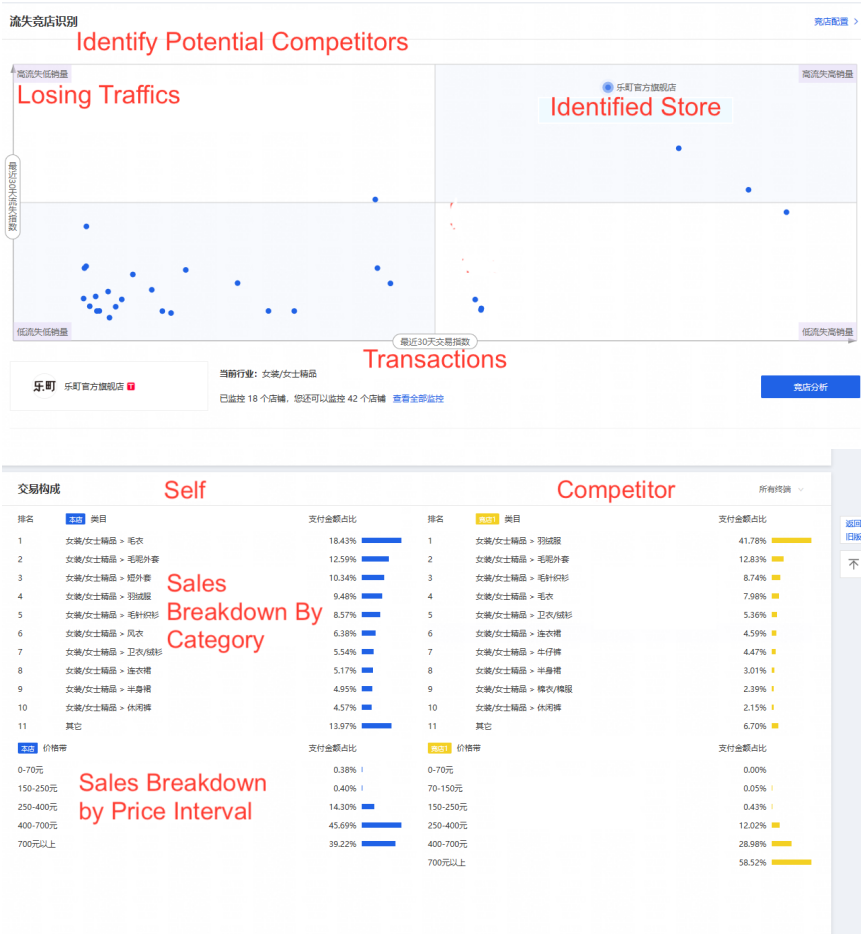

Figure A.3: Traffic Module Interface

Notes: This is a screenshot that illustrates the web interfaces of the paid competition module in the data analytics tool.

Figure A.4: Adoption and Price Change of Market (Pro) Module

*Notes:*number of orders placed for the Market Pro module. Red line depicts number of orders placed by first time users, while the colored area graph differentiate renewal purchases based on the number of renewal. Number of users are re-scaled.

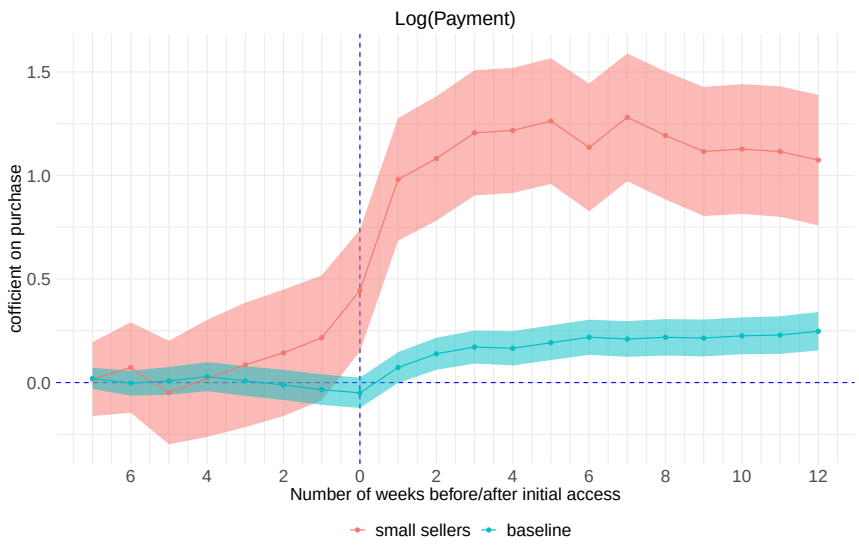

Figure A.5: Effects of Acquiring Own Store Traffic and Customer Data on Log Sales

Notes: the graph shows coefficient γ_k for $k = -11, ..., 11$ where $k = 0$ is the week of adoption, separately for large and small stores regarding adoption of traffic module. A store is small if week sales at week $t - 12$ is in the bottom quarter. Dependent variable is log sales in the previous week. All regressions include controls for stores characteristics, performance and actions profile in the pre-adoption period.

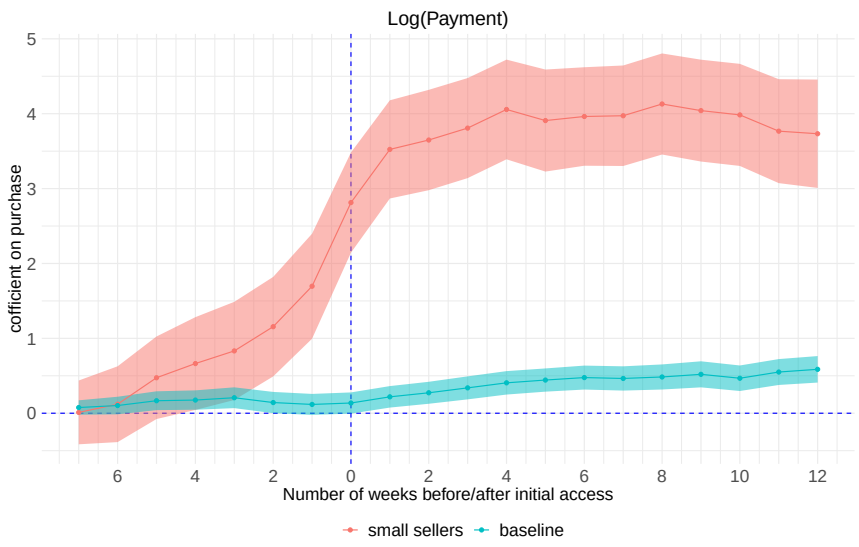

Figure A.6: Effects of Acquiring Top-Ranked stores and Products Data on Log Sales

Notes: the graph shows coefficient γ_k for $k = -11, ..., 11$ where $k = 0$ is the week of adoption, separately for large and small stores regarding adoption of Market (Pro) module. A store is small if week sales at week $t - 12$ is in the bottom quarter. Dependent variable is log sales in the previous week. All regressions include controls for stores characteristics, performance and actions profile in the pre-adoption period.

Table A.1: Sellers' Revenues Using Free Trial

	Dependent variable:				
	Daily Revenues				
	Raw	Log	Winsor 99th	Winsor 99.5th	Winsor 99.9th
	(1)	(2)	(3)	(4)	(5)
Free Trial	39,636.580	1.033	3,304.512**	5,797.472***	6,377.569**
	(39,881.150)	(0.792)	(1,499.935)	(1,856.171)	(2,795.305)
Dep Var Mean	5782.28	6.21	3385.25	3817.47	4543.91
Observations	8,809,122	8,809,122	8,809,122	8,809,122	8,809,122
R^2	0.647	0.671	0.676	0.697	0.692
Adjusted R^2	0.638	0.663	0.669	0.690	0.685

Notes: Regressions based on a sample of 220k sellers who are the target of the promotion campaign. Observations are at seller-day level spanning from June 8 to July 20, 2020. Table presents the estimated effect of free trial from the second stage specification of 4. Dependent variables are daily revenues measured in log scale and in levels with different winsorization thresholds. **p<0.05, ***p<0.01.

B Estimation Result

Table B.1: Estimates of Main Parameters

Parameter	Estimates	Std Err
α_0	-9.09e-4	0.17e-4
$\alpha_{\text{(normed) log size}}$	5.609e-4	0.04e-4
$\alpha_{\text{(normed) open time}}$	-1.93e-4	0.01e-4
σ_γ	4.30	0.81
$\sigma_{\text{sales ind}}$	0.27	0.00
$\sigma_{\text{log sales}}$	-0.56	0.00
$\rho_{\gamma,t}$	0.78	0.06
$\theta_{\gamma,k=\text{top rankings,open time}}$	1.25	0.12
$\theta_{\gamma,k=\text{agg demand,open time}}$	1.11	0.03
$\theta_{\gamma,k=\text{competitors,open time}}$	-1.76	0.17
$\xi_{\gamma,k=\text{top rankings}}$	0	-
$\xi_{\gamma,k=\text{agg demand}}$	0.64	0.22
$\xi_{\gamma,k=\text{competitors}}$	1.88	0.15
$\rho_{k:\text{top rankings \& agg demand}}$	0.02	0.05
$\rho_{k:\text{top rankings \& competitors}}$	-0.42	0.16
$\rho_{k:\text{competitors \& agg demand}}$	-0.16	0.07
$\xi_{j=\text{mkt pro}}$	0	-
$\xi_{j=\text{mkt std}}$	-1.97	0.02
$\xi_{j=\text{comp}}$	-1.36	0.00

C Matching Methodologies and Details

The matched samples are constructed using a random sample of about 57,000 stores in industries related to consumer electronics covering the period of April 2017 to December 2018. For each product module, i.e. market (std), market (pro), traffic and competition, matched samples are constructed separately from the same underlying population. For each specific product module, stores are classified into three categories: 1) non-users are stores that have never purchased the module in the sample period; 2) first-time users are stores that have purchased the module for the first time during the sample period and 3) returning users are stores that have purchased the product module in the past that may or may not renewed during the sample period. The analysis on matched samples focus on comparing non-users and first-time users. Therefore, when constructing the matched sample, returning users are excluded from the population. Notice that stores may have different status with respect to the particular product module of interests. That is, some stores might be returning users for traffic module but are first-time users for market (std) module, in which case these stores will be excluded in construction of the traffic matched sample, but will be included for the construction of the market (std) sample.

For each specific product module, the matched sample is constructed in the following steps. First, stores that are first-time users are categorized into different cohorts based on the week t_0 when purchase is made, where $t_0 \in \{2017-12, ..., 2018-40\}$. Exact number of cohorts as well as size of the cohorts vary by module. The most popular module market (std) have 648 qualified first-time users while the least popular module competition only has 158 users. For each cohort, the matched sample is constructed by selecting comparable stores in the non-users sample that are comparable along the following dimensions:

- *basic characteristics*: rating, days opened, type, industry

- *performance*: sales, number of orders, quantity sold, traffics and conversion rates
- *strategies*: frequency of changing product title, adding/removing products

For variables on performance and strategies, all are matched from week $t_0 - 11$ to week t_0. To be specific, the matched sample is constructed by first running the following specification

$$Purchase_i = \alpha + \sum_{t=t_0-11}^{t_0} X_{it}\beta_t + Z_i\gamma + \varepsilon_i$$

where X_{it} include variables on performance and strategies in each particular week t as described above and Z_i includes all time-invariant characteristics. $Purchase_i$ is a dummy that equals 1 if the store purchase the module in week t_0 and equals 0 if stores are non-users. Notice that first-time users are not included for construction of the matched sample except for being used as the target of the match for the cohort they belong to. Given the specification, the matched sample is constructed by selecting from the non-users with highest purchase likelihood. Cohort size of the matched sample is the same as cohort size of the first-time users' sample for the specific cohort t_0. The final matched sample for a product module is consists of all cohort-specific matched sample for that product module. Cohort (t_0) dummies are included in the analysis for the matched sample to control for cohort-specific effects. By construction, first-time users and their matched non-users counterpart should behave similarly along all matched dimensions (performance, strategies) up till week t_0 due to the requirements applied in the matching process.